

Causal Dependence Plots

Project Notes — RAI

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How to use these notes

This document is the working knowledge base for the project, kept separate from `docs/` (the mentor’s source material, which stays unedited so it is always safe to check a note against what was actually handed to us).

Layout

`main.tex` the master file: preamble, macros, TikZ styles, and the `\input` list below.

`sections/` one file per topic, mirroring the handout’s own structure (`docs/CDP_explainer_students.pdf`).
Compile order equals reading order.

`recall.tex` a *separate*, standalone self-quiz document (compiles to its own `recall.pdf`) — questions first, answers at the end, for testing yourself *without* this reference open. Add to it as you read further; don’t fold it into `main.tex`.

Build with `latexmk -pdf main.tex` (or `pdflatex` run twice, so the table of contents and cross-references settle).

Conventions

- **Cross-references:** every section carries a `\label{sec:...}`; refer to it elsewhere with `\ref` or `\hyperref[sec:...]{...}` rather than repeating content.
- **REUSE vs. NEW:** the mentor’s handout tags every idea as applying an existing result or contributing a new one. Kept here as the `\reusetag{Source}` / `\newtag` macros (defined in `main.tex`), rendering as **[REUSE: Source]** / **[NEW]**. This tag is the organizing device for the whole project — keep using it.
- **Cite by section:** when a note is based on the handout, reference its section number (e.g. “handout §1.3”) so it is easy to jump back to the PDF and check nuance.

- **Don't edit docs/**: if a note turns out to be wrong once something is understood better, fix the note, not the memory of the PDF — the PDF is the ground truth handed down from the mentor.

Day to day

1. When a source is read (paper, handout section), add a dated entry to the reading log (§10) — even two lines.
2. Pull any new term into the glossary (§8) immediately, before the exact definition used *in this project* is forgotten (causal-inference terms often carry paper-specific conventions).
3. When a section grows past its own topic, split it into a new file under **sections/** and add an `\input` line in **main.tex** — but don't pre-create empty sections for material not yet read.
4. Anything unresolved goes to the open-questions section (§11) instead of sitting half-understood inside a content section.

Starting scaffold

The content sections were seeded from **docs/CDP_explainer_students.pdf** (§0–§2) and the CDP paper's introduction (arXiv:2303.04209, §1 + start of §2.1). They are **not exhaustive** — most of the arXiv paper's §2 (§2.1–2.6, its own careful methodology section, per the mentor's prescribed reading order) has not been folded in yet, nor has the Loftus “Trustworthy Causal Explanations” slides (now read, see the reading log) or a full pass of the IML PDP chapter. Treat the seeded notes as a skeleton to correct and extend while actually reading, not as a finished summary.

Mentor's prescribed reading order (given 2026-09-19, see the reading log): (1) Molnar's PDP chapter; (2) the CDP paper, arXiv:2303.04209, §1–2 carefully, skim the rest; (3) this project explainer, **docs/CDP_explainer_students.pdf**. Follow that order, not whatever order these notes happen to be organized in.

1 Assumed background

Source: handout preamble (“Assumed background: DAGs, the do-operator, backdoor adjustment, potential outcomes, regression, a little machine learning.”). Everything from §2 onward leans on all six of these.

Running example, used throughout: age $W \rightarrow$ exercise $A \rightarrow$ weight $M \rightarrow$ blood pressure R ; a model f predicts 10-year cardiovascular risk, $\hat{Y} = f(X)$.

1.1 Notation

A quick-reference table for every symbol used in these notes — consult it whenever a formula reads “obviously” but isn't.

Symbol	Meaning
W, A, M, R	The running example's features: age, exercise, weight, blood pressure (in that causal order).

Symbol	Meaning
X	The full feature vector, e.g. (W, A, M, R) ; X_{-A} means “all features except A .”
f	The fitted prediction model being explained (any learner).
$\hat{Y}$	The model’s output, added to the causal graph as a node, $\hat{Y} = f(X)$.
Y^*	The “constructed outcome” $f(X)$ treated as a variable for causal-inference purposes — the same quantity as $\hat{Y}$, written differently depending on whether the point being made is graphical ($\hat{Y}$) or statistical (Y^*).
$P(\cdot), P(dw)$	A probability distribution; $P(dw)$ is shorthand for integrating/summing against the distribution of W (treat it as “weighted by how W is actually distributed in the population”).
$E[\cdot], E_W[\cdot]$	Expectation; a subscript names which variable’s distribution the average is taken over.
$\text{do}(A = a)$	Pearl’s do-operator: <i>set</i> A to a by intervention (delete the arrow into A), as opposed to conditioning on $A = a$ in observational data (see below).
$\perp$	Statistical independence, e.g. $\{R(a)\} \perp A \mid W$ reads “independent of A , once you condition on W .”
$\theta(a)$	The population target of a dependence plot at level a (e.g. the average TDP, Eq. 6).
$\hat{\theta}(a)$	An <i>estimator</i> of $\theta(a)$ from data. A hat over any symbol ($\hat{\mu}$, $\hat{\pi}$, $\hat{\text{se}}$, ...) always means “estimated from a finite sample,” as opposed to the population quantity underneath it.
$\mu(a, z)$	The outcome regression, $E[Y^* \mid A = a, Z = z]$ (§4).
$\pi(a \mid z)$	The propensity/treatment regression, $P(A = a \mid Z = z)$ (§4).
Z	A generic adjustment/covariate set (e.g. the parents W , or a larger valid set) — kept as a separate letter from the running example’s W, M, R so formulas stay general across choices of adjustment set.
$Q(dr \mid a, w)$	The reduced form : the conditional law of the non-parent features given (A, W) (§3).
$\Psi(a)$	The bridge mean $E[Y(A, Z(a))]$ that the NIDP turns out to equal (§3).
$\sqrt{n}$ (rate)	Shorthand for “a regular estimator whose error shrinks like $1/\sqrt{n}$ ” — the standard, non-exotic rate that licenses ordinary $\hat{\theta} \pm 1.96 \hat{\text{se}}$ confidence intervals.
$o_p(n^{-1/2})$	“Shrinks faster than $1/\sqrt{n}$, in probability” — the condition cross-fitting needs the <i>product</i> of two nuisance errors to satisfy (§4).
$g_k(a), B_k, \psi_k$	A bin’s fixed weight function, the bin itself, and the resulting bin target $\psi_k = \int_{B_k} \theta(a) g_k(a) da$ (§5).
K_h, θ_h	A smoothing kernel at bandwidth h , and the resulting smoothed curve $\theta_h = K_h * \theta$ (§5).

Symbol	Meaning
$J(a)$	The interaction remainder in $\theta(a) - B = [\text{NDDP}(a) - B] + [\text{NIDP}(a) - B] + J(a)$ (§2).
Partial- R^2	The fraction of otherwise-unexplained variance an unmeasured confounder U would need to explain to matter — the currency of the sensitivity analysis in §6.

Table 1: Notation used throughout these notes.

Watch for this overload: the handout reuses the letter R for two different things. In the running example, R is *blood pressure*. But in §3’s general identification result, R denotes “the features other than A and its parents W ” — which in the running example is actually the *pair* (M , blood pressure) together, not blood pressure alone (weight M is also a non-parent of A). Read R as “blood pressure” in running-example prose, and as “the generic non-parent remainder” inside reduced-form formulas — the handout doesn’t distinguish them notationally, so context has to.

1.2 DAGs (directed acyclic graphs)

Nodes are variables, arrows are *direct* causal effects (relative to the other variables drawn), and “acyclic” means no sequence of arrows loops back to its start. The **ECM** (explanatory causal model) in this project is exactly such a graph over the model’s features — an assumption brought to the explanation, not something derived from f itself.

Vocabulary this project actually uses:

- **Parents** of A : nodes with an arrow directly into A (here, just W).
- **Descendants** of A : everything reachable by following arrows *out* of A (here, M , R , $\hat{Y}$).
- **Non-descendants** of A : everything else (here, W , plus anything upstream of or unrelated to A) — the exact set §4’s covariate-selection result is about.
- **Backdoor path**: a path between A and the outcome that starts with an arrow *into* A (i.e. runs through a common cause) — the confounding path adjustment needs to block.

One caution worth internalizing early: conditioning on the wrong node can *create* bias rather than remove it (classically, conditioning on a **collider** — a node two arrowheads point into). This is why “valid adjustment set” is a precise graph-theoretic notion (Pearl’s backdoor criterion, next), not just “control for everything measured.”

1.3 The do-operator

Notation $\text{do}(A = a)$: *set* A to a by intervention, as opposed to *observing* $A = a$ in data that happened to be collected. The distinction matters whenever A shares a common cause with the outcome:

- $P(R \mid A = a)$ — among people who happen to exercise at level a (contaminated by whatever else predicts exercising at that level, e.g. age).
- $P(R \mid \text{do}(A = a))$ — if everyone were *forced* to exercise at level a (severs whatever normally determines A , so no contamination).

Graphically, $\text{do}(A = a)$ means: delete every arrow *into* A (its value no longer depends on its natural causes), then let everything else propagate forward as usual. This is exactly what the TDP diagram in Figure 2 (§2) draws as a severed edge — TDP and PCDP are literally this operation applied to the ECM.

1.4 Backdoor adjustment

What confounding actually is. Age has *two separate* effects here: it affects blood pressure directly (general aging, nothing to do with exercise), and it also affects how much people exercise (for whatever reason — older people might exercise more or less than younger people). Because age drives *both* exercise and blood pressure, a naive comparison of blood pressure between high- and low-exercisers picks up two things tangled together: (1) whatever blood pressure difference exercise itself actually causes, and (2) a spurious difference that’s really just age showing up twice — once through its direct effect on blood pressure, once through its effect on who ends up exercising how much. That second, spurious piece is **confounding**: age *confounds* the naive exercise–blood-pressure comparison. Graphically, it’s exactly a **backdoor path** (§1.2 above): the path $\text{exercise} \leftarrow \text{age} \rightarrow \text{blood pressure}$ connects exercise and blood pressure without exercise having to cause anything, simply because they share a cause.

Why we bother with the marginal-averaging trick. We don’t want “the correlation between exercise and blood pressure among whoever happens to be at each exercise level” (that’s contaminated by age, per above) — we want “what blood pressure would look like if the *same* population, with the *same* age makeup, were compared at different exercise levels.” Forcing every exercise level to be evaluated against the identical, population-wide age distribution (the $P(W = w)$ in Eq. 1 below) is precisely how you strip out age’s contribution and isolate exercise’s own effect. This is *why* we do the adjustment at all, not just how.

The bridge from an interventional question to something estimable from observational data. **Backdoor criterion**: a covariate set W is *valid for adjustment* if (a) it contains no descendant of A , and (b) it blocks every backdoor path from A to the outcome.

What “valid” buys you, and what goes wrong without it. “Valid” means exactly this: plug W into the formula below, and you get the *correct* causal quantity $P(R \mid \text{do}(A = a))$ — not just some number, the right one. Each half of the criterion guards against a specific way of getting a *wrong* number instead:

- **Violating (b)** — a backdoor path left unblocked — means some confounding is still in the room. If there were a second confounder, say socioeconomic status S , that also affects both exercise and blood pressure, and you adjusted only for age (not S), the path $\text{exercise} \leftarrow S \rightarrow \text{blood pressure}$ would still be open. The “adjusted” number would still be contaminated by S , exactly as in the naive calculation above, just by a different amount.
- **Violating (a)** — including a descendant of A — can go wrong two ways. If the descendant is something A causes on the way to the outcome (a *mediator*, like weight M , when what you want is the *total* effect), conditioning on it blocks part of the very effect you’re trying to measure — you’d end up computing something closer to a direct effect than a total one. If the descendant is a **collider** (something caused by both A and something else), conditioning on it can manufacture a spurious association that wasn’t there to begin with.

When both (a) and (b) hold, the formula

$$P(R \mid \text{do}(A = a)) = \sum_w P(R \mid A = a, W = w) P(W = w) \quad (1)$$

is *guaranteed* to equal the true interventional quantity — that guarantee is the entire content of “valid.”

In words: to get blood pressure’s distribution under a forced exercise level a , go age group by age group (w), look at blood pressure among the people who actually exercised at level a within that age group ($P(R \mid A = a, W = w)$), and then combine those age-group numbers into one overall number by weighting each age group by how common it actually is in the population ($P(W = w)$) — not by how common it is among people who exercise at level a .

The crucial move: average over the **marginal** distribution of W (the actual population mix), not the distribution of W *among people at $A = a$* — that second thing is what a naive $E[R \mid A = a]$ implicitly does, and it is exactly where confounding sneaks back in.

Worked toy version of the running example. Say age is 50% young / 50% old *in the population overall* — i.e. combining the high-exercise and low-exercise groups together, not within each group separately. (In real data the age mix *inside* the high-exercise group could easily be, say, 70% old / 30% young if older people tend to exercise more — that mismatch between each group’s own age mix and the population’s age mix is precisely the confounding the adjustment below is designed to remove.) And (invented purely to illustrate) blood pressure is as in Table 2.

	high exercise	low exercise
young	110	120
old	130	150

Table 2: Toy blood-pressure cell means by age and exercise level.

Adjusted:

$$\text{high exercise : } 0.5(110) + 0.5(130) = 120$$

$$\text{low exercise : } 0.5(120) + 0.5(150) = 135$$

In words: for “high exercise,” take the blood pressure of a young high-exerciser (110) and the blood pressure of an old high-exerciser (130), and average them *using the population’s actual 50/50 age split* — not whatever age split happens to show up among the high-exercisers in the data. Do the same for “low exercise.” This gives a clean -15 causal contrast, because both numbers are averaged over the *same* 50/50 age split. A naive $E[BP \mid A=\text{high}]$ vs. $E[BP \mid A=\text{low}]$ would instead average each group over *its own* age mix (however age happens to correlate with exercise in the data).

Making the naive number concrete. Suppose that, because older people happen to exercise less in this data, the high-exercise group is actually 80% young / 20% old, and the low-exercise group is 20% young / 80% old (this is the confounding: each group’s own age mix differs from the population’s 50/50). The naive, unadjusted comparison uses *each group’s own* age mix:

$$\text{naive } E[BP \mid A=\text{high}] = 0.8(110) + 0.2(130) = 114$$

$$\text{naive } E[BP \mid A=\text{low}] = 0.2(120) + 0.8(150) = 144$$

giving a naive contrast of $114 - 144 = -30$ — *twice as large* as the true -15 causal effect computed above. The extra -15 is pure confounding: high-exercisers look even better than they really are (via exercise) partly because they also skew younger, and younger people have lower blood pressure regardless of how much they exercise. Adjustment (forcing both groups to be evaluated at the same 50/50 age split) removes exactly that extra piece, leaving only exercise’s own effect.

Why parents specifically? The set of all parents of A always satisfies the backdoor criterion: any path into A must pass through one of A ’s parents immediately before reaching it, so conditioning on all parents blocks every such path. That is why the handout’s core formula ($\theta(a) = E_W[E[Y^* | A = a, W]]$, $W = \text{parents of } A$; see §2) can use *just* the parents — and why §4 can then ask whether a *larger* valid set (all non-descendants) does better on variance, without needing a different formula.

One distinction worth keeping crisp: **valid is not the same as best**. Multiple different sets can each be valid (each plugs into Eq. 1 and gives the exact correct number) while differing in how *precisely* that number can be estimated from a finite sample — that’s the “efficient set” question in §4, a separate question from validity.

1.5 Potential outcomes

An equivalent (Neyman–Rubin) way of stating the same thing in terms of per-person counterfactuals rather than graph surgery. For person i , posit $R_i(a) =$ “the blood pressure person i would have had if their exercise had been set to a ” — only the value at their *actual* A_i is ever observed; the rest are counterfactual.

Assumptions this needs (same substance as the backdoor criterion, different language):

- **Consistency:** $R_i(A_i) = R_i$ — the potential outcome at the observed treatment equals what was actually seen.
- **Ignorability / unconfoundedness:** $\{R(a)\} \perp A | W$ — treatment is “as good as randomly assigned” once conditioned on W .
- **Positivity / overlap:** every stratum of W has some chance of any A -level occurring in the data, $0 < P(A = a | W = w) < 1$ — what §6’s “non-overlap ratio” diagnostic checks.

Why this project needs it beyond the simple $R(a)$: the **NIDP** is a “bridge mean” $\Psi(a) = E[Y(A, Z(a))]$ — a *nested* potential outcome, mixing a person’s actual A with the downstream features Z set to where they would be under a hypothetical $A = a$. That is genuinely potential-outcomes language, not something the do-operator alone writes down as easily — see §3.

1.6 Regression

The tool that actually gets fit to data once identification says *what* to estimate. Two regressions do essentially all the work in §4:

- **Outcome regression** $\mu(a, z) = E[Y^* | A = a, Z = z]$.
- **Propensity / treatment regression** $\pi(a | z) = P(A = a | Z = z)$.

AIPW combines both (and is right if *either* is estimated well). “Regression” here can mean anything from OLS to a random forest — which is exactly why cross-fitting (train/evaluate on separate splits) is needed once flexible, ML-grade regressions are used for $\hat{\mu}$ and $\hat{\pi}$ and a valid confidence interval is still wanted.

1.7 A little machine learning

f is “any learner” — linear model, random forest, neural net, whatever was fit to predict cardiovascular risk. This project treats f as a **black box**, queried only by feeding it synthetic inputs and reading off predictions (never inspecting its internals) — that is what makes CDPs “model-agnostic.”

Two ML facts turn into project rules, not just background color:

- Flexible learners can nearly reproduce their own training outcomes (a forest is a classic case) — this is *why* §7 insists explanatory curves are drawn on data f never trained on.
- A learner that **splits on** A (e.g. a tree) can make the true dependence curve θ itself jumpy/discontinuous in a — this is *why* §5 treats the smooth-curve estimator as needing an extra smoothness assumption a jumpy θ would violate.

Open questions

- Work the toy backdoor-adjustment arithmetic above with actual made-up individual-level data (not just cell means) to see the confounded vs. adjusted contrast come out of a regression fit, not a hand calculation.
- Pin down precisely how positivity/overlap (potential-outcomes language) and the graph’s positivity condition (§4: “the population positivity condition doesn’t get harder” when adjusting on more covariates) are the same statement in two vocabularies.

2 The big picture: why PDPs aren’t enough, and what CDPs do instead

Source: handout §0; CDP paper (arXiv:2303.04209) §1 intro. Status: skeleton from a first read — reconcile with the paper’s own §1–2 (read carefully, per the mentor’s prescribed order) once read in full. New to DAGs / the do-operator / backdoor adjustment / potential outcomes? Start with §1 — everything below assumes it.

2.1 The problem

A fitted model f (any learner) exists, and a picture of how its output depends on one feature A is wanted, when the features are *causally related* to each other (not independent).

Running example (used throughout — reuse it in notes and questions so everything lines up): age $W \rightarrow$ exercise $A \rightarrow$ weight $M \rightarrow$ blood pressure R ; f predicts 10-year cardiovascular risk.

2.2 The PDP, and why it isn’t causally valid

PDP [REUSE: Friedman 2001]: for each a , set everyone’s exercise to a , *keep other features as recorded*, average predictions. Problem: it silently holds weight and blood pressure fixed — not what would actually happen if someone exercised more. It answers “what if only exercise changed and nothing downstream reacted,” rarely the causal question actually wanted.

Formally (Molnar’s IML book, ch. 19, following [REUSE: Friedman 2001]): split all the features the model uses into the one(s) being plotted, x_S , and everything else, x_C (S and C partition every

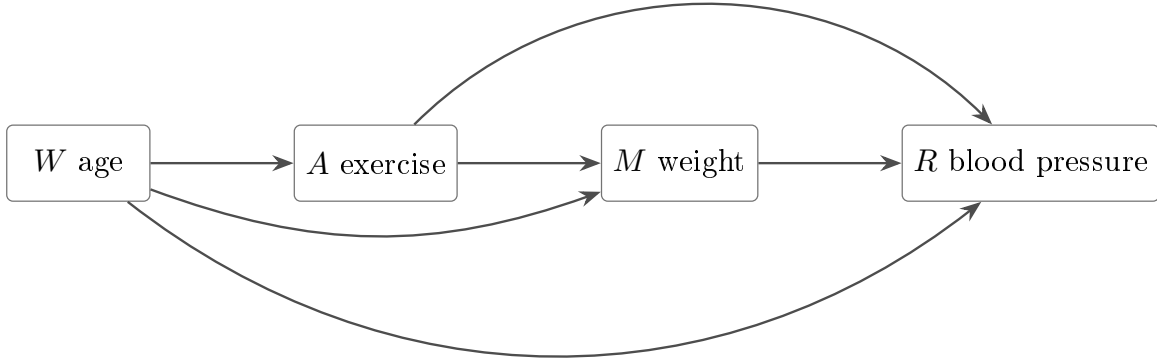


Figure 1: The running example’s full causal graph (ECM), reproduced from handout §0: every earlier feature is drawn as a direct cause of every later one (age $\rightarrow$ exercise, weight, blood pressure; exercise $\rightarrow$ weight, blood pressure; weight $\rightarrow$ blood pressure).

feature f takes as input). The partial dependence function is

$$\hat{f}_S(x_S) = \mathbb{E}_{X_C}[\hat{f}(x_S, X_C)] = \int \hat{f}(x_S, X_C) dP(X_C). \quad (2)$$

In words: freeze the feature(s) you care about at the value x_S you want to plot on the x -axis, then average the model’s prediction over every possible value the *other* features X_C could take, weighted by how often that combination actually occurs ($dP(X_C)$ is the real-world distribution of X_C). That gives one number for that x_S ; repeating it across a range of x_S traces out the curve.

Why an integral, not a sum? — the math one level down. $\mathbb{E}_{X_C}[\cdot]$ always means “probability-weighted average,” but what that average looks like as a formula depends on whether X_C is discrete or continuous.

If X_C could only take finitely many values $c_1, \dots, c_K$ (say, weight rounded to the nearest kilogram), the weighted average would be an ordinary sum:

$$\mathbb{E}_{X_C}[\hat{f}(x_S, X_C)] = \sum_{k=1}^K \hat{f}(x_S, c_k) P(X_C = c_k).$$

But weight and blood pressure are *continuous* — they can take any real value on a range, so there are infinitely many possible values, and the probability of hitting any *one exact* value is 0 (e.g. $P(\text{weight} = 71.3428\dots \text{kg exactly}) = 0$; only *intervals* of values have positive probability). “Multiply by $P(X_C = c_k)$ and sum” breaks down here — every term would be $0 \times (\text{something})$.

The fix is the same one ordinary calculus uses to define area under a curve. Probability for a continuous variable is described by a **density** $p(x_C)$, where $P(a \leq X_C \leq b) = \int_a^b p(x_C) dx_C$ (a density is a *rate* — probability per unit of x_C — not a probability itself, which is why it’s allowed to exceed 1). Chop the range of X_C into thin slices of width Δ ; on each slice X_C is approximately discrete, landing in that slice with probability $\approx p(x_C)\Delta$, so the weighted average over all the slices is

$$\sum_k \hat{f}(x_S, x_C^{(k)}) p(x_C^{(k)}) \Delta \xrightarrow{\Delta \rightarrow 0} \int \hat{f}(x_S, x_C) p(x_C) dx_C.$$

That limit — summing over infinitely many, infinitely thin slices — is exactly what an integral sign *means*. It’s the same idea as “area under a curve is the limit of a sum of thin rectangles” from a first calculus course, just with $\hat{f}(x_S, x_C) p(x_C)$ playing the role of the curve’s height.

Why $dP(X_C)$ instead of $p(x_C) dx_C$? Writing $dP(X_C)$ is a deliberate generalization: it means “weighted by however probability is actually spread out,” without committing to whether X_C is discrete, continuous, or a mix. When X_C is continuous, $dP(X_C) = p(x_C) dx_C$; when it’s discrete, $\int(\cdot) dP(X_C)$ collapses back into the ordinary sum $\sum_k(\cdot) P(X_C=c_k)$ above — one symbol covers both cases, which is why $P(dw)$ in the notation table (§1.1) got the same “integrate or sum, whichever applies” description.

Where did $p(x_C^{(k)}) \Delta$ go in the Monte Carlo sum below? Eq. (3) is $\frac{1}{n} \sum_i \hat{f}(x_S, x_C^{(i)})$ — no density, no Δ , every row weighted equally by $1/n$. That’s not a different formula from the slice-sum above; the density weight is still there, just hidden inside *which points you’re summing over*. Here’s the derivation, one step at a time.

1. Go back to the fixed grid of slices from before, spaced Δ apart, with representative points $x_C^{(1)}, x_C^{(2)}, \dots$ — these are evenly-spaced, chosen *by you*, regardless of where real data lives. That’s why each one needed an explicit $p(x_C^{(k)}) \Delta$ multiplier: a grid point in a crowded region of the population and a grid point in a near-empty region get exactly one term each on the same grid, so probability has to be bolted on by hand to tell them apart.
2. Now overlay your actual dataset on that same grid. Let N_k be the number of your n rows whose $x_C^{(i)}$ lands in slice k . Group the raw (unweighted) sum by which slice each row falls into — $\hat{f}$ barely changes across one thin slice, so every row in slice k contributes almost the same value, $\hat{f}(x_S, x_C^{(k)})$:

$$\sum_{i=1}^n \hat{f}(x_S, x_C^{(i)}) \approx \sum_k N_k \hat{f}(x_S, x_C^{(k)}).$$

3. Divide both sides by n :

$$\frac{1}{n} \sum_{i=1}^n \hat{f}(x_S, x_C^{(i)}) \approx \sum_k \underbrace{\frac{N_k}{n}}_{\text{share of your data in slice } k} \hat{f}(x_S, x_C^{(k)}).$$

4. This is the step that answers the question. N_k/n is the *empirical* fraction of your dataset that happens to land in slice k . If your dataset rows are a representative sample of the population (each one drawn from the true distribution of X_C), the Law of Large Numbers says that fraction converges, as n grows, to the slice’s *true* probability — which is exactly what a density means:

$$\frac{N_k}{n} \xrightarrow{n \rightarrow \infty} P(X_C \in \text{slice } k) \approx p(x_C^{(k)}) \Delta.$$

So $p(x_C^{(k)}) \Delta$ didn’t disappear — it’s sitting inside N_k/n . A crowded region of the population (high p) automatically contributes *many* rows to your dataset, so it automatically gets many terms in the plain, unweighted sum; a rare region contributes few rows and few terms. The multiplying-by-density step and the counting-more-rows-there step are doing the identical job — multiplying explicitly *and* sampling proportionally to density would double-count the same weighting. Concretely: 10 people in your data with weight $\approx 70\text{kg}$ contribute 10 near-identical terms to the raw average, automatically

giving that weight bracket 10× the say of a bracket only 1 person has — that *is* the density weighting, just paid for in row-count instead of in an explicit multiplier. (This relies on the dataset actually being representative of the population you want the curve to describe — the same “which reference population” issue flagged in §7.)

Putting the two limits together — $\Delta \rightarrow 0$ (arbitrarily fine slices) and $n \rightarrow \infty$ (arbitrarily large data) — turns the slice-sum into the exact integral, which is why Eq. (3) is called an *estimator* of Eq. (2) rather than a separate definition: same target, approached by letting real data stand in for the unknown true distribution.

You can’t literally integrate over the true, unknown distribution of X_C , so in practice $\hat{f}_S$ is estimated by **Monte Carlo averaging** over the dataset you actually have:

$$\hat{f}_S(x_S) = \frac{1}{n} \sum_{i=1}^n \hat{f}(x_S, x_C^{(i)}). \quad (3)$$

In words: go through every row i in your dataset, one at a time. For each row, swap in the fixed value x_S for the feature(s) you’re plotting, but keep that *same row’s own, real, observed* values for every other feature ($x_C^{(i)}$ — the “ (i) ” means “as recorded for individual i ,” not a made-up value). Run the model once per row under this swap, then average all n of the resulting predictions. “Monte Carlo” here just means: instead of doing the exact integral in Eq. (2) (impossible, since you don’t know the true distribution $P(X_C)$), approximate it by an average over samples you actually have — your dataset itself stands in for that unknown distribution, and the approximation gets better the more rows you average over.

This is exactly the same computation as averaging every individual’s ICE (individual conditional expectation) curve at that x_S — a PDP curve *is*, by construction, the pointwise mean of all the ICE curves in the dataset. And it’s precisely the operation already described above in words (“set everyone’s exercise to a , keep other features as recorded, average predictions”): Eq. (3) is that sentence written as a formula, with $S = \{A\}$ (exercise) and $C = \{M, R\}$ (weight, blood pressure) in the running example.

A fully worked example, with numbers. Suppose the dataset has just $n = 5$ people (a toy size, purely so every row can be written out by hand), and a toy model

$$\hat{f}(A, M, R) = 100 - 2A + 0.3M + 0.4R$$

predicting a cardiovascular risk score from (exercise hrs/week A , weight kg M , blood pressure R) — a made-up linear formula, invented only so the arithmetic below is checkable by hand; a real $\hat{f}$ would be some fitted ML model with no such closed form.

i	A_i (real)	M_i (real)	R_i (real)
1	2	70	120
2	5	65	110
3	0	90	140
4	3	75	125
5	1	85	135

Table 3: Toy dataset, $n = 5$ people, each with their own recorded (A_i, M_i, R_i) .

To compute $\hat{f}_S(5)$ — “what would average predicted risk be if *everyone* exercised 5 hrs/week” — Eq. (3) says: take each row in Table 3, replace A_i with the fixed value 5, keep that same row’s own M_i, R_i completely untouched, run $\hat{f}$, then average all 5 results:

i	$(x_S, x_C^{(i)}) = (5, M_i, R_i)$	$\hat{f}(5, M_i, R_i)$
1	(5, 70, 120)	$100 - 10 + 21 + 48 = 159.0$
2	(5, 65, 110)	$100 - 10 + 19.5 + 44 = 153.5$
3	(5, 90, 140)	$100 - 10 + 27 + 56 = 173.0$
4	(5, 75, 125)	$100 - 10 + 22.5 + 50 = 162.5$
5	(5, 85, 135)	$100 - 10 + 25.5 + 54 = 169.5$

Table 4: The 5 terms actually being averaged in Eq. (3) at $x_S = 5$. Column 1 (exercise) is the same fixed 5 in every row — the “freeze x_S ” part. Weight and blood pressure differ row by row — each person’s own, untouched $x_C^{(i)}$.

$$\hat{f}_S(5) = \frac{159.0 + 153.5 + 173.0 + 162.5 + 169.5}{5} = \frac{817.5}{5} = 163.5.$$

In words: we are averaging over **5 numbers, one per person in the dataset** — not over 5 possible values of exercise, and not over some abstract population. Person 3, who in reality never exercises at all ($A_3 = 0$, Table 3), still contributes a term to this average: their *predicted risk in a pretend world where they exercised 5 hrs/week but kept their actual weight and blood pressure* (173.0 — the single highest of the five, driven entirely by their real $M_3 = 90, R_3 = 140$, since A is fixed at 5 for every row here, exercise contributes nothing to the differences between rows). Repeating the same 5 rows with $x_S = 0$ instead of 5 gives $\hat{f}_S(0) = 173.5$ instead — a different final number, because only column 1 changed while the 5 real (M_i, R_i) pairs stayed exactly the same. That’s the entire mechanism behind *one point* of the PDP curve; repeating this 5-row average once per value of a traces out the whole curve.

The independence assumption, made concrete. Eq. (3) pairs the fixed $x_S = a$ with *every* row’s own $x_C^{(i)}$, including rows whose real exercise level was nothing like a . That’s fine only if exercise and (weight, blood pressure) are unrelated — but they aren’t (that’s the whole running example). Concretely, in Table 4: Person 3 really never exercises ($A_3 = 0$) and really carries a high weight/BP profile ($M_3 = 90, R_3 = 140$) — plausibly *because* they don’t exercise. Eq. (3) still uses their (M_3, R_3) as a stand-in for “someone who exercises 5 hrs/week,” as if that weight/BP combination were a realistic thing to see at high exercise, when in this population it may never actually occur there. This is the same mechanism as the confounding problem in §1 — it’s why the PDP “isn’t causally valid” named in this subsection’s title, and it’s exactly what the CDP framework below is built to fix.

Aside: PDP-based feature importance [REUSE: Greenwell, Boehmke & McCarthy 2018]. Not part of the CDP framework itself, but built directly on the PDP curve just defined, and worth having on hand since “does this curve actually vary, or is it flat?” is a question that comes back repeatedly later.

The idea in one sentence: a flat PDP curve means the feature barely changes the prediction (not important); a curve that swings up and down a lot means the feature matters a lot (important). Feature importance turns “how much does the curve vary” into a single number.

Numerical features. Evaluate the PDP curve $\hat{f}_S$ at its K *unique observed values* $x_S^{(1)}, \dots, x_S^{(K)}$ (literally every distinct value the feature actually takes in the dataset, not an arbitrary grid), then measure how spread out those K curve heights are around their own average:

$$I(x_S) = \frac{1}{K-1} \sum_{k=1}^K \left(\hat{f}_S(x_S^{(k)}) - \frac{1}{K} \sum_{k=1}^K \hat{f}_S(x_S^{(k)}) \right)^2. \quad (4)$$

In words: the inner term $\frac{1}{K} \sum_k \hat{f}_S(x_S^{(k)})$ is just the *average height* of the curve across its K points — the flat reference line you’d see if the feature had zero effect. For each point, subtract that average and square the result: a point on a flat curve sits close to the average, so this is close to 0; a point far from the average (the curve is swinging there) gives a large squared number. Add up all K squared deviations and divide by $K-1$. This is nothing more exotic than the ordinary **sample variance formula** ($s^2 = \frac{1}{n-1} \sum (y_i - \bar{y})^2$, same $K-1$ denominator for the same reason) applied to the K curve heights instead of K raw data points — “importance” here literally *means* “variance of the PDP curve.”

Worked example, continuing the toy dataset above. Since $\hat{f}_S(a) = 100 - 2a + 73.5 = 173.5 - 2a$ in the running toy (the average of $0.3M_i + 0.4R_i$ over the 5 people works out to 73.5), the $K = 5$ unique real exercise values $\{0, 1, 2, 3, 5\}$ give curve heights $\{173.5, 171.5, 169.5, 167.5, 163.5\}$, averaging to 169.1. Squared deviations: $4.4^2, 2.4^2, 0.4^2, (-1.6)^2, (-5.6)^2 = 19.36, 5.76, 0.16, 2.56, 31.36$, summing to 59.2. So $I(A) = 59.2/4 = 14.8$. If exercise instead had *no* effect on $\hat{f}$ at all, all 5 curve heights would be identical, every squared deviation would be 0, and $I(A)$ would come out exactly 0 — that’s the whole point of the measure.

Categorical features. Categories don’t sit on a number line, so “deviation from the mean” isn’t quite the natural notion; a cruder, more robust measure is used instead — the **range rule**:

$$I(X_S) = \frac{\max_k (\hat{f}_S(x_S^{(k)})) - \min_k (\hat{f}_S(x_S^{(k)}))}{4}. \quad (5)$$

In words: take the PDP’s highest value across the K categories, subtract its lowest value, divide by 4. *Why divide by 4?* A rule of thumb borrowed from the normal distribution: about 95% of a normal distribution’s mass sits within ± 2 standard deviations of its mean — a band 4 standard deviations wide. So if all you know is a variable’s range (max–min) and want a rough guess at its standard deviation, range/4 is a reasonable back-of-envelope estimate (range $\approx 4\sigma \Rightarrow \sigma \approx \text{range}/4$). It’s deliberately crude and typically *underestimates* the true spread — but with only a handful of categories (K small), there’s little basis for a real variance estimate anyway, which is exactly why the cruder range-based version replaces Eq. (4) here.

Worked example (toy, illustrative numbers, following Molnar’s own season/bike-rental example): PDP-predicted bike rentals by season, Spring=150, Summer=180, Fall=165, Winter=90. Range = $180 - 90 = 90$, so $I(\text{season}) = 90/4 = 22.5$.

Two things this measure is *not*. (1) It only sees the feature’s *main effect* — the shape of its own PDP curve — and is blind to *interactions*: a feature that matters only by interacting with another feature (e.g. flipping sign depending on a second feature’s value, the heterogeneous-effects problem from the recall questions above) can have a nearly flat, low- I PDP curve while still being genuinely important to the model — permutation feature importance would catch that; this measure would miss it. (2) It weights every *unique value* equally, regardless of how many rows share it — a value only one person in the dataset has counts exactly as much as a value shared by hundreds, unlike the

PDP curve itself (Eq. (3)), which already weights by how common a value’s neighborhood is via the Monte Carlo average.

2.3 The fix: use the ECM to define “changing A ”

ECM = “explanatory causal model” — the analyst’s DAG over the *features*. It is an assumption brought to the explanation, not a claim verified by the model. Given an ECM, there is more than one way to ask “how does output depend on A ” — there are (at least) four, depending on whether downstream features are allowed to respond:

Plot	Question	In the running example
TDP (total)	set $A = a$; let everything downstream respond	weight and BP move
NDDP (natural direct)	set $A = a$; hold downstream features at their <i>natural</i> values	this is exactly the PDP (hand-out’s Thm. 2.10)
NIDP (natural indirect)	keep your own A ; move downstream features to where they’d be under $A = a$	your exercise, but the weight/BP you’d have at exercise a
PCDP (partially controlled)	set $A = a$ <i>and</i> fix another feature	exercise = a , weight = m

Table 5: The four named dependence plots.

The genuinely new fact worth internalizing early: **NDDP = PDP**. The plot everyone already uses is a special case of this framework — the “direct effect only” member of the family, not a separate method.

2.4 Constructed outcome — the trick that makes this identifiable

Treat the model’s output as a variable: $Y^* = f(X)$. Then the **average TDP is a plain back-door adjustment**:

$$\theta(a) = E[f(X) \mid \text{do}(A = a)] = E_W[E[Y^* \mid A = a, W]], \quad W = \text{parents of } A. \quad (6)$$

In words: the average model output you’d get from forcing everyone to exercise level a ($\theta(a)$, left-hand side) equals a two-step average on the right: first, within each age group w , average the model’s prediction over the people who actually exercise at level a (inner $E[Y^* \mid A = a, W]$); then average those age-group numbers together using the real population mix of ages ($E_W[\cdot]$) — the same “average over the marginal, not the conditional” move as Eq. 1, now applied to the model’s output instead of a raw outcome. **[REUSE: Pearl 2009, Thm. 3.2.2; Zhao & Hastie 2021 (theorem)]**; **[NEW]** is what this implies specifically for CDPs. The equations for weight and blood pressure are never needed: $\text{do}(A = a)$ leaves the parents’ distribution alone and leaves every downstream conditional given (a, w) alone. This is *why* §3 can get away with only needing the parents plus a “reduced form,” not the full fitted ECM.

2.5 The picture that organizes the whole family

[REUSE: Shpitser & Tchetgen Tchetgen 2016] Add the model’s output $\hat{Y}$ to the graph as a child of every feature. Then TDP/PCDP are ordinary node interventions ($\text{do}(A = a)$, $\text{do}(A =$

$a, C = c$)), while NDDP/NIDP are **edge** interventions (set a on the arrow into $\hat{Y}$ only, vs. on the arrows into the child features only). Normally, one node “acting two ways at once” needs cross-world assumptions data cannot check — but here $\hat{Y} = f(X)$ is a *known function*, so nothing cross-world is actually invoked. That is the structural reason NDDP reduces to an ordinary average (the PDP) and NIDP reduces to an ordinary single-world quantity (§3).

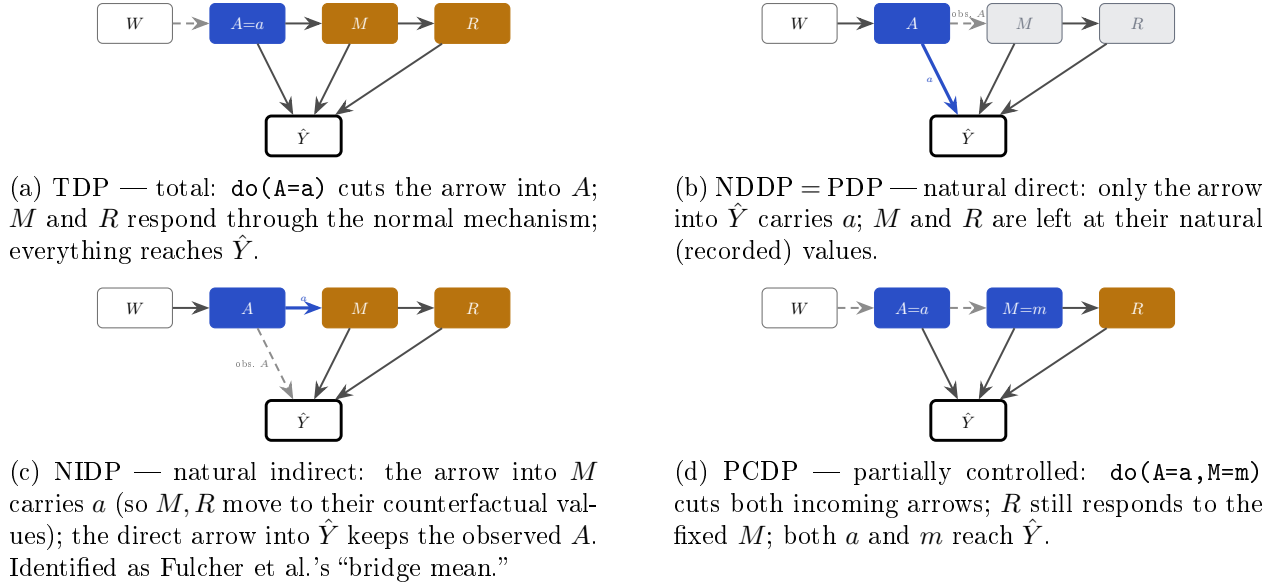


Figure 2: The four dependence plots as different interventions on the same explanatory causal model. *Simplified for legibility*: the full ECM (Figure 1, handout §0) also has direct edges $W \rightarrow M$, $W \rightarrow R$, $A \rightarrow R$ among the features, and — since $\hat{Y}$ is a child of every feature — a direct $W \rightarrow \hat{Y}$ edge; all four are omitted here so the direct/indirect mechanism stays readable. Blue = set to the counterfactual value; amber = responds/recomputed; gray = kept at its natural (recorded) value; dashed = edge severed, or carrying the natural value instead of a .

Legend:

- set to the counterfactual value
- responds / recomputed
- kept at natural (recorded) value
- active edge (solid)
- - - severed edge, or an edge carrying the natural value instead of a

Direct + indirect $\neq$ total. [NEW], modest.

$$\theta(a) - B = [\text{NDDP}(a) - B] + [\text{NIDP}(a) - B] + J(a),$$

In words: take how far the total curve is from the overall average prediction ($\theta(a) - B$, left side). You might expect that gap to split cleanly into “how much of it is the direct effect” plus “how much of it is the indirect effect” (the two bracketed terms on the right). It doesn’t, in general — there’s a leftover piece $J(a)$ that has to be added to make the two sides match, and $J(a)$ is exactly zero only in the special case where f treats A and the downstream features additively (no interaction between them).

Here B is the average prediction and J is an interaction remainder that vanishes only when f is additive in A and the downstream features. Toy counterexample: $A \sim \text{Bern}(p)$, $R(a) = a$, $f = a \cdot r$; the total contrast is 1, but the direct and indirect contrasts are each only p . **Don't assume direct + indirect sums to total** — J should be plotted alongside them, not ignored.

Open questions

- Work through why exactly the parents' distribution is untouched by $\text{do}(A = a)$ — is this just the definition of “parents,” or does it need the ECM to have no unmeasured confounding among features?
- Read arXiv §1–2 in full (the mentor's prescribed order has this *before* rereading the handout) for the paper's own formal definitions, then compare notation to the handout's.

3 Identification: what each plot needs from the data

Source: handout §1.1. This is the intern-term paper's (“Paper A”) first result. Status: skeleton from a first read.

3.1 The reduced form

Let R be the features other than A and its parents W . Let $Q(dr \mid a, w)$ be the conditional law of R given (A, W) — the **reduced form**. Crucially, this is *learnable straight from data* — no ECM structural equations needed.

3.2 Per-plot identification table

Plot	Population target	What it consumes	Statistical character
TDP	$\int P(dw) \int Q(dr \mid a, w) f(a, w, r)$	parents + reduced form	AIPW; $\sqrt{n}$ at discrete levels
NDDP PDP	= $E[f(a, X_{-A})]$	no causal nuisance at all — only f and the reference distribution of X_{-A}	sample mean of a known function; $\sqrt{n}$ at each a
NIDP	$\int P(da', dw) \int Q(dr \mid a, w) f(a', w, r)$	parents + reduced form + law of A given W	its own influence function; $\sqrt{n}$ at discrete levels
PCDP	needs the graph <i>inside</i> R	more than the reduced form (two graphs with the same data give different answers)	g-formula

Table 6: Identification of each named plot (handout §1.1).

Reading the three dense formulas in words: TDP's $\int P(dw) \int Q(dr \mid a, w) f(a, w, r)$ says — go age group by age group ($\int P(dw)$, weighting by how common each age group actually is); within each

age group, average over what weight-and-blood-pressure values typically go with exercise level a there ($\int Q(dr \mid a, w)$, the reduced form); then plug each (a, w, r) combination into the model f and average. NIDP’s formula is almost identical but starts by also averaging over exercise levels a' drawn the way they actually occur for each age group ($\int P(da', dw)$) before asking what downstream values go with the *target* level a — that’s the “your own exercise, but their downstream values” mixing. NDDP’s $E[f(a, X_{-A})]$ is the simplest: just plug the fixed value a in for exercise, leave every other feature exactly as recorded in the data, run it through f , and average — no age groups, no reduced form, nothing learned about causal structure at all.

Takeaway to hold onto: three of the four named plots (TDP, NDDP, NIDP) need *only* the parents of A plus one reduced form — fitting every equation of the ECM is never required. Only PCDP needs the internal graph structure of R . This is what makes “adjustment” (§4) a real alternative to “propagation” (§6) for three of the four plots.

3.3 NIDP = a known bridge mean (reuse, not a new discovery)

[REUSE: Fulcher, Shpitser, Marealle & Tchetgen Tchetgen 2020, JRSSB] “Your own A , but the downstream values you’d have had under $A = a$ ” is exactly their **bridge mean** $\Psi(a) = E[Y(A, Z(a))]$ from the population-intervention indirect-effect literature (Z = downstream features, $Y = f$) — in words, plug each person’s *own actual* exercise level (A , not a) into the model, but pair it with the downstream feature values ($Z(a)$) they *would have had* under the hypothetical exercise level a , then average over everyone. It mixes one real, observed input with one counterfactual input for the same person. Their Lemma 1 gives the formula, their Theorem 1 gives its influence function (one term drops out here because $Y^* = f(X)$ is *deterministic*, not noisy), and their doubly-robust estimator applies directly.

What’s **[NEW]** here specifically: recognizing the published NIDP *is* this object, that the covariate set reduces to just the parents of A , and the simplifications that follow from f being a known function rather than an unknown outcome-generating process. Caveat: handled for **discrete A only** this term — continuous- A NIDP needs its own theory.

3.4 Direct + indirect $\neq$ total

[NEW], modest. With B the average prediction:

$$\theta(a) - B = [\text{NDDP}(a) - B] + [\text{NIDP}(a) - B] + J(a),$$

(in words: the total curve’s distance from the average, on the left, isn’t simply the direct effect’s distance plus the indirect effect’s distance — there’s a leftover term $J(a)$ needed to make the two sides balance; see §2 for the full walkthrough). J is an interaction remainder — it vanishes only when f is additive in a and the downstream features. Toy counterexample: $A \sim \text{Bern}(p)$, $R(a) = a$, $f = a \cdot r$: the total contrast is 1, but the direct and indirect contrasts are each only p . **Don’t assume direct + indirect sums to total** — J should be plotted alongside them, not ignored.

Open questions

- Why does PCDP specifically need the graph *inside* R , when TDP/NDDP/NIDP don’t need any equations at all? (Intuition: fixing an intermediate feature “cuts” a path in a way sensitive to which downstream variable causes which — worth confirming with a toy SCM.)
- Verify §1.1’s table by exact enumeration in toy SCMs — handout §1.9 marks this as intern-feasible / light supervision (• intern-feasible). Good first task.

4 Estimation: AIPW, cross-fitting, and which covariates to adjust for

Source: *handout §1.2–1.3. Status: skeleton from a first read.*

4.1 Estimating one level, discrete A (§1.2, mostly REUSE)

Three classical estimators (Robins/Rotnitzky/Zhao 1994; Hahn 1998): regression (regress Y^* on (A, Z) , average), inverse-propensity weighting, or **AIPW**, which combines both and is *doubly robust* — consistent if *either* piece is right:

$$\hat{\theta}(a) = \frac{1}{n} \sum_{i=1}^n \left[\frac{\mathbf{1}(A_i = a)}{\hat{\pi}(a | Z_i)} (Y_i^* - \hat{\mu}(a, Z_i)) + \hat{\mu}(a, Z_i) \right]. \quad (7)$$

In words, person by person: start with the regression’s best guess for person i ’s outcome if they’d had exercise level a , given their covariates ($\hat{\mu}(a, Z_i)$ — this term alone is used for everyone). Then, *only for the people who actually did exercise at level a* ($\mathbf{1}(A_i = a)$ switches this next part on or off), add a correction: how far their real outcome was from that regression guess ($Y_i^* - \hat{\mu}(a, Z_i)$), inflated by how rare it was for someone with their covariates to end up at exercise level a ($1/\hat{\pi}(a | Z_i)$ — rarer means a bigger correction, since that person is doing more work standing in for others like them). Average this person-by-person quantity over everyone, and that’s $\hat{\theta}(a)$. The “doubly robust” part: if $\hat{\mu}$ is wrong but $\hat{\pi}$ is right, the correction term fixes it on average; if $\hat{\pi}$ is wrong but $\hat{\mu}$ is right, the correction term is small anyway because the regression guess was already good.

Cross-fitting [REUSE: Chernozhukov et al. 2018]: fit $\hat{\mu}, \hat{\pi}$ on one data half, evaluate on the other, swap. This lets flexible ML be used for the nuisance functions while still getting a valid Wald interval $\hat{\theta}(a) \pm 1.96 \widehat{\text{se}}$, *provided* the two nuisance errors shrink fast enough *jointly* (their product must be $o_p(n^{-1/2})$). Cross-fitting removes a technical condition, not the underlying accuracy requirement — a common misreading to watch for.

4.2 Which covariates Z to adjust for (§1.3)

[REUSE: Rotnitzky & Smucler 2020], [NEW] application. The parents W are a *valid* adjustment set but not necessarily the lowest-variance one. Adding non-descendants of A that help predict Y^* lowers variance without changing the answer (same logic as good covariates in a randomized trial).

Key result: when f uses all features, the set of *all non-descendants of A* is never worse than any other valid set in large samples (among the adjustment estimators used here). Two practical facts:

- The *population* positivity condition does not get harder — because $P(a | \text{non-descendants}) = P(a | W)$ by the graph. What gets harder is *fitting* a bigger regression in finite samples. So: **fit the propensity on W , fit the outcome regression on the larger set.**
- This result uses only the graph’s *arrows* — never its equations. (The handout notes this was verified exhaustively on small graphs in the review — a good sanity-check exercise to redo.)

Open questions

- Work through *why* adding non-descendants can only help (not hurt) asymptotic variance — is there a clean intuition beyond “more relevant predictors of Y^* reduce residual variance”?

- Handout §1.9 marks “cross-fitted AIPW at discrete levels; coverage on oracle testbed” as intern-feasible / moderate supervision (• moderate supervision), and “parents vs. efficient set (variance drop)” likewise — plausible first coding tasks once toy-SCM verification (§3) is done.

5 Continuous A : bins first, curves second

Source: handout §1.4. Status: skeleton from a first read.

5.1 Bins (primary target)

At each bin B_k , report the fixed-weight average $\psi_k = \int_{B_k} \theta(a) g_k(a) da$ — in words, don’t report the true curve $\theta(a)$ at every single value of a in the bin; instead, report one number per bin, obtained by averaging $\theta(a)$ over all the a -values inside that bin (B_k), using a weighting rule g_k that’s written down *before* looking at the data (e.g. “just weight every a -value in the bin equally”). Why fix the weight ahead of time rather than use the observed distribution of A inside the bin: weighting by the observed distribution makes the target depend on the data law and adds a term to the standard error. This is a regular, $\sqrt{n}$ target, estimated by the same AIPW machinery (Eq. 7) with weight $g_k(A)/\hat{\pi}(A | Z)$ — and it does not depend on which valid Z was chosen (§4).

5.2 Smoothed curve — two tiers

- **Smoothed curve (extension, supervisor-owned):** $\theta_h = K_h * \theta$ at a declared bandwidth h — in words, the smoothed curve’s value at a point is a local weighted average of the true curve θ near that point, where the kernel K_h is the rule for how much weight nearby a -values get (points closer than the bandwidth h count more, points farther away count less or not at all) — kernel-weighted AIPW, bands via multiplier bootstrap. Works for *any* f — important because a tree that splits on A can make θ itself jumpy, and the “smooth curve” theory below assumes smoothness a jumpy θ doesn’t have.
- **Smooth curve (secondary, if θ is actually smooth):** cite Takatsu & Westling’s debiased local-linear estimator, or Semenova & Chernozhukov’s cross-fitted series estimator (its variance has an extra term that must not be dropped). Pseudo-outcome must be Kennedy’s construction — the naive “ $\hat{\mu} + (Y^* - \hat{\mu})/\hat{\pi}$ regressed on A ” estimates the wrong thing (handout notes it was checked to be off by 0.9 in a toy example — a concrete trap to avoid reinventing).

5.3 Bias policy

[NEW] correction. Any uncorrected smoother at its best bandwidth has bias as large as its standard error, so naive confidence bands **under-cover**. The project’s earlier prototype hit exactly this (0.89 coverage instead of ~ 0.95). Fix: debiasing, deliberate undersmoothing, or report bands for θ_h itself rather than pretending they are bands for the unsmoothed θ .

Open questions

- Handout marks the (primary) fixed-bin estimator as intern-feasible (• moderate supervision — “representer, bootstrap over a finite vector”). The smoothed curve is explicitly • supervisor-owned / not intern-feasible — don’t invest here without direction.
- Understand precisely why a data-dependent bin weight adds a standard-error term — worth a small derivation.

6 Adjustment vs. propagation, and sensitivity analysis

Source: *handout §1.5–1.6. Status: skeleton from a first read.*

6.1 Adjustment vs. propagation (§1.5)

[NEW] study. Four arms compared: parents only; the efficient covariate set (§4); an estimator exploiting the DAG’s factorization; and full **parametric propagation** (fit every ECM structural equation, simulate — what the *original* CDP paper’s method does).

Findings to reproduce/understand, not just quote:

- A confounder living entirely **downstream** (e.g. between weight and blood pressure, independent of exercise) does *not* invalidate the parent-adjustment formula for the *total* curve — the formula is right whether or not that confounder is drawn in the graph. It changes the data (and so the curve), but adjustment still recovers the right curve for whatever the data is. Propagation only inherits this if its fitted conditionals match the observational ones — which holds *exactly* for linear OLS (an algebraic identity: fitted path coefficients always compose to the reduced-form slope) but can fail for nonlinear f : a Gaussian propagation model can match the mean and variance exactly and *still* get $E[f]$ wrong (handout cites a checked example: error 0.047 with mean and variance exact). **Nonlinearity is where propagation’s extra assumptions can bite.**
- When the ECM’s parametric equations are correct, propagation can be *more precise* and can extrapolate outside the region of data overlap — but that should be declared as model-based extrapolation, not treated as free. Overlap is diagnosed via the **non-overlap ratio** **[REUSE: Bao & Schomaker 2026]**.
- Whether exploiting the DAG’s factorization can beat plain adjustment at all is answered by **[REUSE: Rotnitzky & Smucler 2020, Thm. 19]**.

6.2 Sensitivity analysis (§1.6)

Mostly **[REUSE: Cinelli & Hazlett 2020; nonparametric bounds: Chernozhukov, Cinelli, Newey, Sharma & Syrgkanis]**, **[NEW]** framing. If an unmeasured U affects A *and* a downstream feature that matters for f , it can bias the adjusted curve (special cases: the bias cancels, or U turns out irrelevant to f). Uses **partial- R^2 sensitivity** to ask: how strong would U need to be to flip the **slope** (never the level — that depends on an arbitrary zero point), flatten the curve, or cross a threshold?

Project-specific (**[NEW]**) content: confounding parameterized *per ECM edge*, with an exact map to the standard Cinelli–Hazlett coordinates (the “tipping strength” is their robustness-value formula, just reparameterized); bounds at each bin; benchmarking against observed predictors; and cross-checking against the §1.5 invariance result above.

Open questions

- Handout marks “propagation arm via DoWhy + invariance demonstrations” as intern-feasible / moderate (● moderate supervision) — a plausible task once §3/§4 basics are solid.
- “Robustness section: linear closed forms, planted confounders, tipping, benchmarking” is intern-feasible / moderate (● moderate supervision), with binned Riesz bounds “only if time allows” — i.e. a stretch goal, not baseline scope.

- Work the toy linear-OLS “path coefficients compose to the reduced-form slope” identity by hand for the running example graph.

7 Practical rules, and what this term’s paper does *not* claim

Source: *handout §1.7–1.9, §2. Status: skeleton from a first read.*

7.1 Two cheap but important rules (§1.7)

[NEW], cheap, important.

1. **Explanatory data must be held out.** $Y_i^* = f(X_i)$ on rows used to *train* f is not a fair sample of what f does (a random forest nearly reproduces its own training outcomes). Draw every curve on rows the model never saw. Out-of-bag / cross-fitted predictions are predictions of *different* models, not of f itself — don’t substitute them.
2. **Name the reference population.** The outer average is over some distribution of W ; training cohort, held-out sample, and deployment population give genuinely different curves whenever the effect varies with w . This term’s inference is all for the *held-out sample’s* population. Carrying a curve to a different population (“transport”) needs extra data and different formulas — explicitly **out of scope** this term.

7.2 What Paper A does *not* claim (§1.8)

Not claiming: that parent adjustment, AIPW, cross-fitting, or dose-response bands are new ideas; that the method works “without a causal model” (it always needs the ECM); that this is the first semiparametric inference for causal explainability (cf. Zhang & Gao 2026; Gao & Zhao 2024 — different quantities, worth knowing about but not the same target); bands for *individual per-person* curves (out of scope — only aggregate curves here).

Why this matters for us as interns: knowing the boundary of the claim keeps write-ups honest and stops accidental overclaiming of novelty that belongs to cited work — exactly what the REUSE/NEW tagging throughout the handout is for.

7.3 Paper C — later project, not this term (§2)

A theory paper making the §2 picture rigorous: why a *deterministic* output node removes the cross-world content of edge interventions; influence functions for all four plots (NIDP’s is Fulcher’s; PCDP’s is a g-formula one); efficiency results when the whole DAG is used; and **graph uncertainty** — from observational data one often cannot orient every edge, so there are several candidate parent sets and several candidate curves. Caution flagged in the handout: for linear-Gaussian models, all graphs in a Markov-equivalence class have identical likelihood, so a BIC “neighbourhood” of orientations is *empty* there; for nonlinear models, orientations actually get scored differently. A couple of intern-sized pieces exist here (simulating candidate curves with IDA software; a merging algorithm) but the core theory is research-level — **not this term’s scope**, noted here mainly so this term’s work doesn’t accidentally drift into it.

Open questions

- None yet — revisit once Paper A’s early tasks (§3–§4) are underway and it becomes clearer where the term’s actual scope will land.

8 Glossary

Definitions as used *in this project* (handout docs/CDP_explainer_students.pdf and Loftus, Bynum & Hansen, arXiv:2303.04209). Each row points to the section where the term is developed.

Term	Means, here	See
ECM	“Explanatory causal model” — the analyst’s DAG over the model’s <i>features</i> (not a claim about the real world, just the assumption used to define the explanation)	§2
Y^* , constructed outcome	The fitted model’s output, $Y^* = f(X)$, treated as if it were an outcome variable for causal-inference purposes	§2
PDP	Partial Dependence Plot (Friedman 2001): average f over the data with feature A set to a , other features left at recorded values	§2
ICE	Individual Conditional Expectation plot: one PDP-style curve per observation, not averaged	§9
PDP-based feature importance	Variance (numerical features) or range/4 (categorical) of the PDP curve’s own values — how much the curve moves, as a single number; blind to interaction effects	§2
TDP	Total Dependence Plot: $\text{do}(A = a)$, let downstream features respond	§2
NDDP	Natural Direct Dependence Plot: set $A = a$, hold downstream features at natural values — provably equals the PDP	§2
NIDP	Natural Indirect Dependence Plot: keep your own A , but downstream features move to where they’d be under $A = a$	§2
PCDP	Partially Controlled Dependence Plot: $\text{do}(A = a)$ <i>and</i> fix another feature	§2
Reduced form, $Q(dr a, w)$	Conditional law of the non-parent features R given (A, W) — learnable straight from data, no ECM equations needed	§3
Bridge mean	Fulcher et al.’s object $\Psi(a) = E[Y(A, Z(a))]$; turns out to be exactly the NIDP	§3
AIPW	Augmented inverse-probability weighting: doubly-robust estimator, consistent if <i>either</i> the propensity or the outcome regression is right	§4
Cross-fitting	Fit nuisance models on one data split, evaluate on another, swap — lets flexible ML be used for nuisances and still get valid Wald intervals	§4

Term	Means, here	See
Efficient covariate set	Not just a valid adjustment set (e.g. parents W) but the one minimizing variance — here, all non-descendants of A when f uses every feature	§4
Non-overlap ratio	Diagnostic (Bao & Schomaker 2026) for how much the curve at a relies on extrapolation vs. observed data overlap	§6
Tipping strength	Cinelli–Hazlett robustness value, reparameterized per ECM edge: how strong an unmeasured confounder U would need to be to flip the curve’s slope	§6
Propagation	Alternative to adjustment: fit every ECM structural equation and simulate counterfactuals person-by-person (what the original CDP paper does)	§6

Table 7: Project glossary.

Add rows as new project-specific terms are met — don’t let this go stale.

9 Bibliography / reuse tracker

One row per paper the handout cites. *Role* is what it is REUSEd for in *this* project. *Read?* should be updated honestly as each is actually worked through — don’t mark done from the handout’s summary alone.

Paper	Role in this project	Used in	Read?
Loftus, Bynum & Hansen — “Causal Dependence Plots” (arXiv:2303.04209)	The source paper. Defines TDP/NDDP/NIDP/PCDP, does full parametric propagation, no sampling-based bands	§2	intro + §1–2.1 only
Friedman (2001)	Original PDP	§2	no (via Molnar’s IML chapter instead — docs/IML/)
Pearl (2009), Thm. 3.2.2	Back-door adjustment theorem, applied to $Y^* = f(X)$	§2	no
Zhao & Hastie (2021)	Supports the back-door-for- Y^* argument	§2	no
Shpitser & Tchetgen Tchetgen (2016)	Edge interventions / the $\hat{Y}$ -augmented-graph picture	§2	no

Paper	Role in this project	Used in	Read?
Fulcher, Shpitser, Marealle & Tchetgen Tchetgen (2020, JRSSB)	NIDP = their “bridge mean” from population-intervention indirect effects; its influence function and doubly-robust estimator	§3	no
Robins, Rotnitzky & Zhao (1994); Hahn (1998)	Classical regression / IPW / AIPW estimators	§4	no
Chernozhukov et al. (2018)	Cross-fitting / double ML	§4	no
Rotnitzky & Smucler (2020)	Efficient covariate-set theory (incl. Thm. 19, factorization vs. adjustment)	§4, §6	no
Takatsu & Westling	Debiased local-linear estimator for smooth dose-response curves	§5	no
Semenova & Chernozhukov	Cross-fitted series estimator (smooth curve)	§5	no
Kennedy et al. (2017)	Correct pseudo-outcome construction for the smooth-curve estimator	§5	no
Bao & Schomaker (2026)	Non-overlap ratio diagnostic	§6	no
Cinelli & Hazlett (2020)	Partial- R^2 sensitivity analysis / robustness value	§6	no
Chernozhukov, Cinelli, Newey, Sharma & Syrgkanis	Nonparametric sensitivity bounds	§6	no
Bonvini et al. (2022); Marmarelis et al. (2023); Jesson et al. (2022); Baitairian et al. (2025)	Sensitivity analysis for dose-response curves generally, instantiated here with $Y^* = f(X)$	§6	no
Zhang & Gao (2026); Gao & Zhao (2024)	Related but distinct semiparametric-inference-for-explainability work — cited to delineate what’s <i>not</i> new here	§7	no
Maathuis, Kalisch & Bühlmann (2009); Perković (2020); Guo & Perković (2021); Chang, Guo & Malinsky (2025); Edgar & Battey (2026); Guo, Perković & Rotnitzky (2023)	Graph-uncertainty / IDA literature — Paper C scope, not this term	§7	no

Table 8: Bibliography and reuse tracker.

Other docs to fold in: docs/Trustworthy Causal Explanations _ Joshua Loftus.docx.pdf (likely broader motivation/framing — not yet mapped to a row above); docs/IML/...Partial Dependence Plot...html (Molnar’s PDP chapter — good PDP intuition before tackling identification).

10 Reading log

Dated, short entries. Not a summary of the source — a record of what was personally taken from it and what was unclear. Newest entry first; add new entries just below this line.

2026-09-19

- Rewrote the whole notes vault from Markdown into this L^AT_EX document with Claude, at the mentor’s/project’s preference for proper math typesetting and a compilable PDF.
- Caught and fixed a real error in Figure 1: it only drew the chain $W \rightarrow A \rightarrow M \rightarrow R$, missing three edges ($W \rightarrow M$, $W \rightarrow R$, $A \rightarrow R$) that the handout’s own diagram has — found by zooming into the source PDF at high resolution and tracing the arrows by hand rather than trusting memory.
- Read docs/Trustworthy Causal Explanations _ Joshua Loftus.docx.pdf in full (only 2 pages). Key facts: this is a Fall 2026, 13-week team project advised by Joshua Loftus (LSE); weekly supervision meetings plus Slack; deliverables include an open-source Python implementation, simulation/real-data experiments, and a report targeting a conference submission (ICML/UAI/NeurIPS) with students as co-authors. It gives an actual week-by-week timeline (Weeks 1–2 foundations, 3–5 discrete AIPW, 6–7 practical rules + interaction plot, 8–9 robustness, 10 break, 11–12 continuous features + comparison study, 13 wrap-up) that lines up closely with handout §1.9’s feasibility table.
- New open question from that doc: Weeks 1–2 ask us to run “the project’s verified check scripts” and reproduce the total-curve identity by exact enumeration using an existing code base — that code base isn’t in docs/. Logged in §11.
- **Mentor gave the actual prescribed reading order** (supersedes my own guessed order below): (1) Molnar’s PDP chapter; (2) the CDP paper, arXiv:2303.04209, **§1–2 carefully, skim the rest**; (3) this project explainer (docs/CDP_explainer_students.pdf). Two corrections to how I’d been scoping things: “§1–2” means the *whole* methodology section (§2.1 through §2.6, not just §2.4–2.6 as I’d guessed), and the paper comes *before* rereading the handout, not after — read the source’s own definitions first, then see how the handout reframes/scopes them for this project, rather than the other way around.
- Finished the “Assumed background” section of these notes (§1). Started step 1 of the prescribed order above: Molnar’s IML PDP chapter (docs/IML/19 Partial Dependence Plot (PDP)...html). Added active-recall questions 39–50 to recall.tex for it as I went.

2026-09-18

- Set up the original notes vault + a concept-map overview artifact with Claude.
- Skimmed (not fully read): docs/CDP_explainer_students.pdf §0–§2 (whole handout structure), docs/2303.04209v2.pdf (CDP paper) §1 intro + start of §2.1 methodology.

- Not yet touched: arXiv paper §2.4–2.6 (the actual CDP definitions/theorems in the source paper’s own notation), docs/Trustworthy Causal Explanations _ Joshua Loftus.docx.pdf, docs/IML/...Partial Dependence Plot...html (Molnar’s PDP chapter — good for building PDP intuition before tackling §1.1’s identification table).
- Open question logged: handout references PLAN_v3.md §10 which isn’t in docs/ — see §11.

11 Questions for the mentor

Running list — bring these to the next meeting, then move answered ones down to §11 (with the answer folded in) instead of leaving them open.

Open

- ☐ docs/CDP_explainer_students.pdf §1.9 says “matches PLAN_v3.md §10” — we don’t have PLAN_v3.md. Could you share it?
- ☐ Confirm which of the §1.9 feasibility-table tasks we’re actually starting with first (the table lists ~8 tasks marked yes/light or yes/moderate).
- ☐ The project brief (docs/Trustworthy Causal Explanations _ Joshua Loftus.docx.pdf) says Weeks 1–2 include running “the project’s verified check scripts” and reproducing the total-curve identity “by exact enumeration in toy models” using an existing, reproducible code base — we don’t have that code base in docs/. Could we get access to the repository?

Answered

(none yet)